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SAE AERO DESIGN 2027 COMPETITION (REGULAR CLASS)

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Abstract/Executive Summary

The Loyola Marymount University SAE Aero Design team is developing a multi-motor, radio-controlled cargo aircraft for the 2027 SAE Aero Design Regular Class Competition. This report presents the team's engineering design, manufacturing plan, and testing strategy. The goal is to design, build, and demonstrate a multi-motor, remote-controlled aircraft that hauls a payload of 2-liter bottles, either empty or filled, while completing the required takeoff, flight course, and landing. This year’s competition imposes constraints on all entries: wingspans between 72 and 96 inches, a maximum gross weight of 55 pounds, a maximum total body length of 120 inches, propulsion via either two or four electric motors, and a 4-cell lithium-polymer battery with a capacity no greater than 2200 mAh. Project success will be measured by the aircraft's ability to complete the flight while carrying the greatest amount of cargo and by the team's ability to accurately predict its best flight performance.

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Chapter I: Introduction

For over 40 years, the SAE Aero Design Competition has been an aircraft challenge, introducing competitors to design and manufacturing under the constraints of a set, predetermined mission. This mission standard replicates industry-level requirements to allow teams to dip their toes into the aerospace industry. Additionally, this event pushes participants to strengthen their ingenuity, project management, documentation, and engineering cycle skills to better prepare for real-world situations. For the regular class, teams create an originally designed remote-controlled aircraft that takes off, completes a flight circuit, and lands successfully while carrying a payload that increases its weight. The competition outlines specific design requirements that must be complied with for any entry to be eligible, as outlined in Chapter IV. With this in mind, the plane must be designed to maximize its carrying capacity while minimizing its functional weight. The flight competition will be held on April 11th in Van Nuys, California.

Chapter II: Problem Statement

According to the SAE Aero Design Rules, “The Regular Class objective is to design a multi-motor cargo aircraft that can maximize mission performance associated with payload delivery. Payload consists of unmodified, commercially available 2L cylindrical plastic bottles, either EMPTY or FILLED, carried on each flight. Successful teams will accurately predict maximum flight score and demonstrate an ability to incrementally increase payload capacity.” The challenge presented with this project is to design, manufacture, test, and compete with an aircraft created from scratch. The aircraft must be able to generate enough lift to lift both itself and empty/filled 2L bottles using either 2 or 4 motors on the wings up to 55lbs total. This must be coupled with controlling the aircraft and managing any offset in center of gravity due to the payload.

Level One Requirements

1. Payload Requirements
    - Regular Class payload shall carry 2L bottles, either empty or filled, carried internally, and properly secured.
2. Weight Limits
    - Aircraft gross take-off weight shall not exceed 55 lbs.
3. Wingspan Limits
    - Regular Class wingspan will be between 72 in and 96 in.
4. Motor Requirements
    - Regular Class will be powered by 2 or 4 motors.
5. Material Prohibitions
    - No FPF, rubber bands, or components to stabilize the aircraft that are not control surfaces.

Chapter III: Background

The SAE Aero Design Competition dates back 40 years and was established to give competitors industry-level experience managing a project from initial standards and requirements to a functioning flight competition. This gives participants a practical way to apply theoretical coursework to a hands-on aerospace event. Competitions are divided into three classes–regular, mini, and advanced.

The regular class is centered around the fundamentals of aircraft design and maximizing a plane's payload capacity, where scoring is primarily based on the additional weight the aircraft can carry while staying under the size and weight specifications. Each year, the rules update to keep the competition constantly changing and encouraging new ideas. This year, there have been multiple changes that include, but are not limited to: smaller wingspan allowances, increasing the number of motors from one to two or four, propeller sizes, payload dimensions, and material restrictions. With these changes in mind, this year's entry must have different designs and priorities as compared to previous years. The wing size and number of propellers demand creativity and ingenuity to conform to new dimensions, including necessary changes to the cargo bay to fit the new bottle payload shapes. Additionally, the scoring for each team has changed compared to previous years. Rather than being focused on maximizing the wingspan of the craft, there is a new target on maximizing the payload capacity.

The competition evaluates teams across three core pillars that mirror aerospace industry standards. These three evaluation metrics are the Flight Performance/Payload Capacity, a Flight Demonstration Readiness Review, and a comprehensive Technical Design Report. During the flight portion, the aircraft must execute specific mission profiles, lift a payload, and take off and land within certain runway length limits. The regular class focuses on fundamental aircraft design, heavy volumetric cargo lifting using 2-liter bottles, multi-motor electric propulsion, and flight score prediction.

We will be using preceding information from the 2022-2023 LMU SAE Aero team to formulate a plan and incorporate design iterations. Namely, the legacy wing foil shape was Eppler 423 due to its high lift-to-drag ratio, making it useful for lifting heavier payload into flight. This legacy design is the starting basis for our wingfoil shape. Additionally, we will be using general aviation planes as a reference due to their purpose as stable, efficient, and simple means for air travel.

Chapter IV: Design Specifications

The specified design requirements are in accordance with the SAE 2027 Regular Class Competition rules. Table 1 is a comprehensive list of all requirements for aircraft parameters.

***Table 1*:** Design Specification Requirements in accordance with 2027 SAE Aerodesign Rulebook [1]

Rule #	Parameter	Requirement	Estimated Capability	Margin	Basis
2.2.1	Aircraft Identification	Identified with school name, mailing address, and email address on the inside or outside.	TBD	TBD	By Design
2.2.2	Aircraft Identification	Team number on aircraft: minimum of 4 in; if vertical surface is smaller, a smaller size is permitted.	TBD	TBD	By Design
2.2.4	Aircraft Identification	University name clearly displayed on wings or fuselage.	TBD	TBD	By Design
2.4.1	CG Requirement and Markings	Shall be flyable at their designated Empty CG position on the submitted 2D aircraft drawing.	TBD	TBD	By Design
2.4.3	CG Requirement and Markings	Markings shall be a minimum of 0.5 inches in diameter centered at the Empty CG position +/-0.25 inches, per the submitted 2D drawings. Flying-wing type aircraft may place the CG markings on the bottom of the wing.	TBD	TBD	By Design
2.5	Gross Weight	Gross take-off weight not to exceed 55lbs.	TBD	TBD	By Design
2.6	Controllability	Aircraft needs to be controllable in flight; If	TBD	TBD	By Design

		equipped with wheeled landing gear, it needs to have a steering mechanism.			
2.8	Radio Control System	Needs to have a 2.4 GHz radio control system with a fail-safe system that reduces throttle to zero immediately.	TBD	TBD	By Design
2.10	Propellers	Use either a spinner or rounded model aircraft type safety nut for propeller mounting; nylon-insert lock nuts prohibited.	TBD	TBD	By Design
2.11	Propellers	Metal propellers are not allowed.	TBD	TBD	By Design
2.12	Material	Lead is prohibited in any part.	TBD	TBD	By Design
2.13	Payload	Payload cannot contribute to the structural integrity of the airframe.	TBD	TBD	By Design
2.19	Power	Batteries need to be commercially available, secured, and free of any hardware or protrusions.	TBD	TBD	By Design
2.22.0	Red Arming Plug	Need to use a discrete and removable red arming plug to arm and disarm the propulsion system; it needs to be integrated into the circuit between the battery and speed controller, located on the positive wire between the battery and speed controller.	TBD	TBD	By Design
2.22.1	Red Arming Plug	Red arming plug: minimum of 9 in away from any propeller; for aircraft with propulsion systems in one	TBD	TBD	By Design

		longitudinal plane, the red arming plug shall be located on the other side of the wing longitudinally; red arming plug located on top near centerline of fuselage of wing external to surface; red arming plug needs to be clearly visible; non-removable portion of arming plug can't have more than one male lead.			
2.23	Receiver System Battery Pack	A separate battery shall be required for the receiver system.	TBD	TBD	By Design
3.8	Take-off	100 ft take-off distance limit; 400 ft distance from initial start before turn.	n/a	n/a	Testing
3.9	Landing Requirements	400 ft landing distance limit.	n/a	n/a	Testing
7.1	Dimension Requirement	Minimum planform wingspan 72 in, maximum planform wingspan 96 in, wing chord greater than 4 in, maximum body axis length less than 120 in.	TBD	Comply	By Design
7.2	Material and Equipment Restrictions	No Fiber-Reinforced Plastic (FRP) (duct tape) on any part of aircraft; exceptions include commercially available FRP motor mount, propeller, landing gear, and control linkage components.	TBD	Comply	By Design
7.3	Propulsion System Requirements	Either 2 or 4 Electric Motors; no restrictions on make or model; Gearboxes, belt drive systems, and propeller shaft extensions are allowed if a one-to-one propeller-to-motor RPM is maintained; Powered by a	TBD	Comply	By Design

		commercially available 4-cell (14.8V) Lithium-Polymer battery pack; max requirement 2200 mAh; Max propeller diameter per motor is 12 in if two motors are used and 9 in if 4 motors are used.			
7.4.0.0	Payload Requirements	Consists of unmodified commercially available 2L cylindrical plastic bottles, carried internally.	TBD	Comply	By Design
7.4.1.1	Payload Requirements	Contain a minimum of 1 bottle during flight attempt	TBD	Comply	By Design
7.4.1.2	Payload Requirements	Bottle Cargo has two configs: EMPTY BOTTLE: Contains unpressurized air or inert material such that bottle weight is between 1-4lbs. FILLED BOTTLE: Contains unpressurized air or inert material such that bottle weight is greater than or equal to 4lbs.	TBD	Comply	By Design
7.4.1.3	Payload Requirements	After a scored flight, the next attempt will increase payload by adding another empty bottle and/or replacing an empty bottle with a filled one.	TBD	Comply	By Design
7.4.2.0	Payload Requirements	Fully enclosed cargo bay: shall not be exposed to airstream at any point, cargo not move or shift during flight, no restriction on size or shape of cargo bay, can have multiple cargo bays	TBD	Comply	By Design
7.5	Payload	To complete a successful	TBD	Comply	By Design

	Unloading	flight for score, post-flight Bottle Cargo unloading shall be accomplished within one minute.			
7.6	Class Scoring	$FFS = FinalFlightScore = (FS + FS + FS)/3 + PPB$	TBD	Comply	By Design

Chapter V: Design Alternatives and Evaluation

This chapter discusses the major design decisions involved in developing a fully functional aircraft. To make the overall design process more manageable, the aircraft was divided into several smaller design components. The primary components considered were the fuselage, wings, tail, control surfaces, landing gear, and power systems. Each component has different design requirements and factors that must be considered. This chapter explores these considerations, along with the advantages and disadvantages of different design options, to help determine the most suitable choices for this project.

The design philosophy for the fuselage was to build a casing for the interior components that the fuselage conceals. Therefore, its design is centered establishing a layout for the payload within the cargo bay first and foremost. There were two different designs that were processed, including a flat layout and a stacked layout. In both designs, there was allocation for a maximum of 4 bottles. Other decisions were made in regards to constraining the payload and choosing door dimensions.

The flat layout would allow all bottles to lay completely flat in the cargo bay, reducing the height of the fuselage and allowing us to move the bottles to any location we want to properly balance the center of gravity. In this iteration, the bottles will be strapped down by Velcro and should be easy to access via a back door. A sketch can be seen in Figure 1.The other layout for the cargo bay is shown in Figure 2. Its goal is to have the center of gravity easily stay in the desired location, no matter how many bottles are loaded into the cargo bay. It will have two shelves in an “X” shape that allows each bottle to rest in their proper space. However, this significantly increases the height of the fuselage, which would result in an extreme door length to maintain a low upsweep angle. Increasing this induces both parasitic drag and form separation [2].

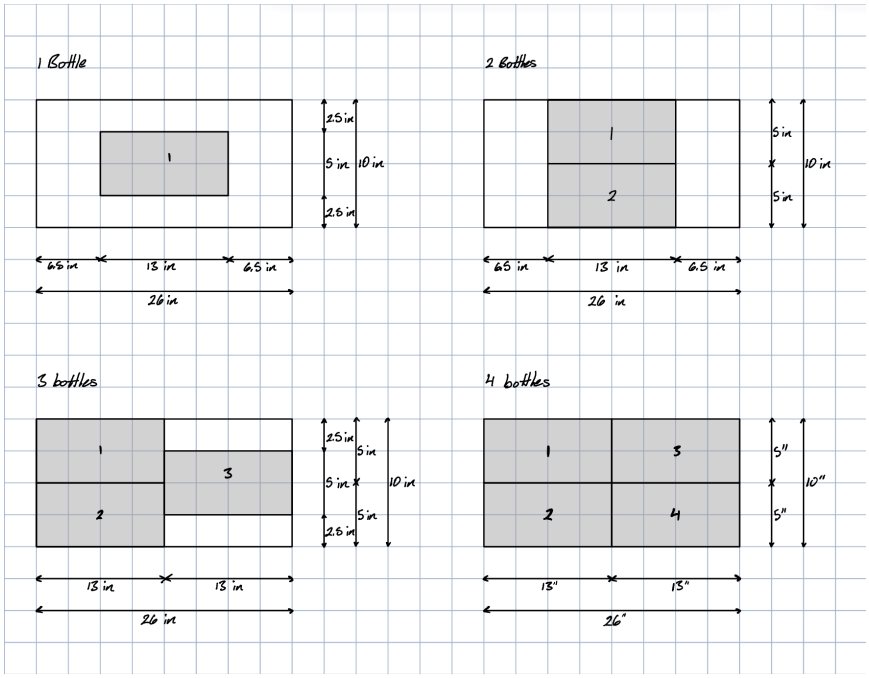


Figure 1: *Flat layout for the cargo bay*

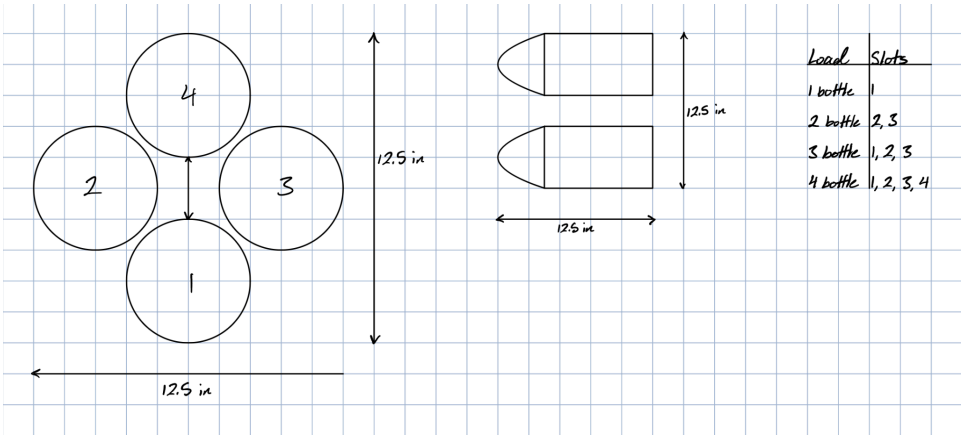


Figure 2: Stacked layout for the cargo bay

Another design challenge that accompanied the addition of payload is properly securing them during flight while ensuring that they can be replaced within 60 seconds. There were two options that were weighed. The first was a quick-drop clamp for the center of the bottle that could be moved around. This would be an easy way for the bottles to be easily fastened and moving the bottles in and out of the clamps' grip would be efficient. But the clamps themselves would prove a challenge to move around quickly. Additionally, we could run into issues if we wanted the clamps to be perfect fits for the bottles, which exact dimensions we are merely estimating. Our other option is Velcro straps that would hold the payload down to the bottom of the cargo bay. Each strap will slide through loops on the cargo floor. These loops will already be built into the cargo bay floor, making it easy to assemble various payload configurations exactly as we want. This solution was chosen for its ease to integrate and work with.

As mentioned, the door has to be set at a specific orientation to keep the upsweep angle shallow for aerodynamic purposes. To avoid this issue altogether, other methods of accessing the cargo bay were considered. Other doors that were considered were wing gull doors and horizontal sliding doors. The wing gull doors would make the payload very easy to access and allow the back of the fuselage to be any shape. However, due to the wings being placed above the fuselage, this door selection would most likely have interaction with the wings. As a result the horizontal sliding door idea was brought up, where the doors would open laterally. Not only would this avoid the upsweep angle issue, but it would also allow the doors to open without hitting the wings and allow for easy payload access. We ultimately decided to move with a different design because the sliding doors would have been significantly more complicated and difficult to integrate. The back door is simple to work with and does not interfere with other components, at the cost of the payload being slightly less accessible and needing to work around the tail arm.

The wing design is one of the most important design decisions in an aircraft as it affects all other aircraft components. The main purpose of the wing is to maximize lift and to minimize drag, and nose-down pitching moment. Several choices in wing design affect these parameters, and must be considered. The main choices in wing design for this project are: wing vertical location, airfoil selection, aspect ratio, taper ratio.

The first design that's needed to be made for wings is their vertical location in relation to the fuselage's center line. The main four configurations, which can be seen in Figure 3 below, are high wing, mid-wing, low wing, and parasol; for this project only high wing, mid-wing, and low wing were considered.

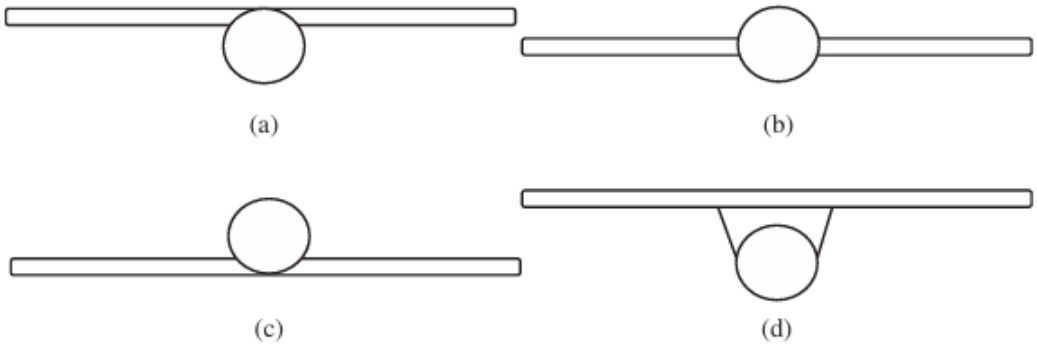


Figure 3: Vertical wing positions: (a) High wing; (b) Mid-wing; (c) Low wing; and (d) Parasol wing [3]

High wing configurations provide greater lateral stability, better low-speed performance, more internal space, better ground clearance, simplified construction, and better lift than mid-wing and low wing. However, they have greater structural weight, more induced drag, and less lateral control than low and mid-win. Low-wing configurations offer better takeoff performance, lower weight, reduced drag, greater lateral control, and improved maneuverability. They however provide less lateral stability, have higher stall speeds, and have longer takeoff

distances. Mid-wing configurations share several features between high and low wing configurations but still have their own pros and cons. They are structurally heavier, more aerodynamically streamlined, have less interference drag than high or low wings, and have a more complex assembly [3]. For this project, a high-wing configuration was determined to be the best choice. The SAE aero requirements involve carrying one or more payloads, making the additional internal space available for cargo storage a significant advantage. The potential for simpler construction, increased ground clearance, and favorable lift characteristics also makes the high-wing configuration well suited for our needs. Although low and mid-wing configurations offer advantages such as reduced drag and improved maneuverability, these characteristics are less important than cargo capacity and ease of construction that a high wing provides.

Airfoil selection is incredibly important in wing design as it is responsible for producing the required lift while decreasing drag and pitching moment. For this project we are choosing from previously designed and published airfoils; considerations made when choosing an airfoil include drag, stall behavior, and structural depth. Airfoil geometry affects airflow and drag. Laminar flow produces less skin-friction drag, and a drag bucket is the range of lift coefficients where drag is low; cruise and climb conditions should fall within this range, an example of this can be seen in Figure 4. Airfoil shape also affects maximum lift and stall behavior. Greater camber increases maximum lift, while sudden flow separation can cause a sharp loss of lift and make the aircraft difficult to control. Therefore, both lift performance and predictable stall behavior are important, desired conditions can be seen in Figure 5. Airfoil thickness also affects structural performance. Thicker airfoils allow for taller wing spars, which can reduce structural weight and improve strength [4].

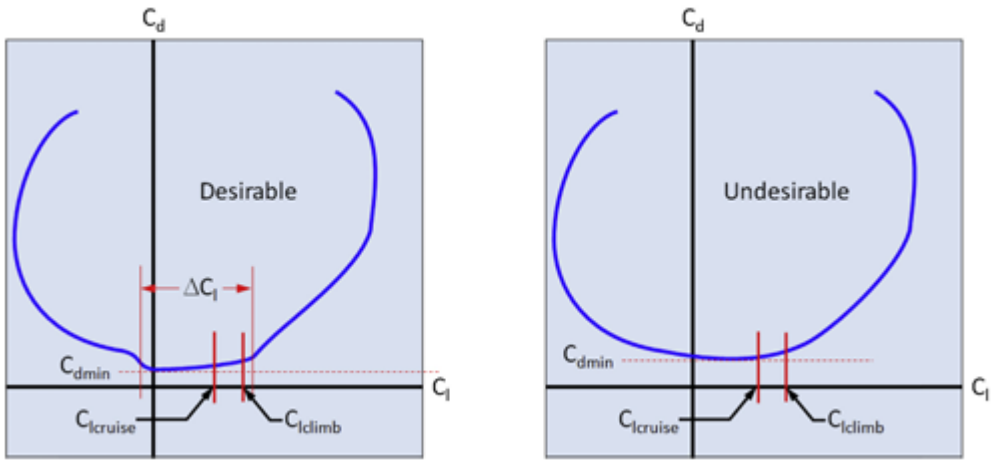


Figure 4: [4]

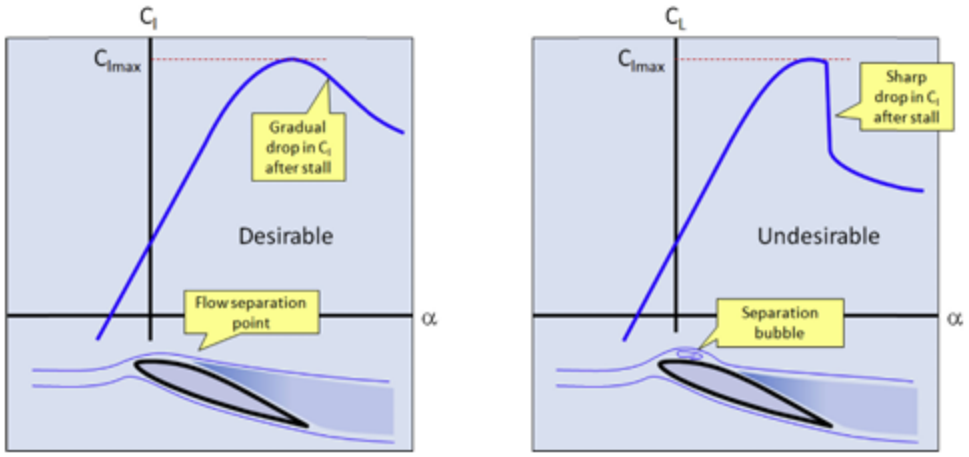


Figure 5: [4]

We researched a few airfoils such as NACA 4415, Seilig s1123, NACA 64(2)-415, NACA 63(2)-215 MOD B, as well as the previous group's airfoil choice, Eppler 423. Ultimately we decided to go for NACA 64(2)-415, due to its thickness, as well as its predictable stall conditions and desired drag bucket at assumed Reynolds number range during flight (200,000-1e6) [5].

The last key design features for a wing are its aspect ratio (AR) and taper ratio (λ). AR is defined as the ratio between the wing span and the chord. As the AR is increased it causes an increase in lift, decrease in drag, more weight, decreased maneuverability, and more difficult manufacturability than a low AR [3]. Despite downsides that come with a high AR, for our uses it would be best to increase it (have longer wings) and have an AR in the 6-8 range. Taper ratio is the ratio between the tip chord and root chord; wings with a $\lambda = 1$ are rectangular; a taper ratio of $0 < \lambda < 1$ is trapezoidal (straight tapered), and a taper ratio of $\lambda = 0$ denotes triangular wings. Wing taper can improve lift distribution, reduce wing weight and bending moment, and improve lateral control. However, it increases manufacturing challenges, and can also affect lateral stability [3]. Furthermore, triangularly tapered wings tend to be an improvement over rectangular wings due to their reduction in induced drag up to 24.4% reduction (under certain conditions) [6]. As mentioned there is an added manufacturing challenge that arises from tapered wings, however due to the significant benefits that come from having tapered wings we decided to go with it over having rectangular wings.

The tail on an aircraft affects its stability, control, and aerodynamic performance, with its main purpose being to maintain stability and provide control while minimizing drag and weight. The main design choices made for the tail were tail configuration, airfoil selection, and aspect ratio.

An aircraft's tail can have several configurations those being; conventional, T-tail, cruciform, H-tail, V-tail, Y-tail, Twin vertical tail, Boom mounted. These configurations can be seen in Figure 6. For our project only conventional, T-tail, and cruciform configurations were considered.

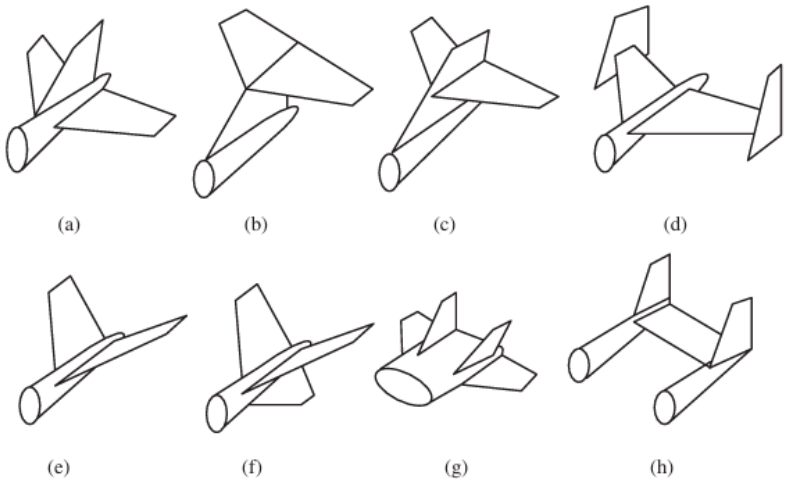


Figure 6: (a) Conventional; (b) T-tail; (c) Cruciform; (d) H-tail; (e) V-tail; (f) Y-tail; (g) Twin vertical tail; (h) Boom mounted [3]

For our project we decided to go with a T-tail configuration; since we are doing a high wing configuration a T-tail aids in preventing wake generated from wings at a high angle of attack, whereas conventional and cruciform wings are affected by this. Another advantage of having a T-tail is that it improves vertical tail efficiency through an end-plate effect, allowing a smaller vertical tail; however, one of the downsides of this design choice is that it requires a stronger and heavier structure to support the horizontal stabilizer [7].

Tails, like wings, share several design choices when it comes to deciding upon an airfoil as well as its AR, however there are a few differences. For tails both vertical and horizontal tails must use a symmetrical airfoil, and the AR for tails (horizontal tail) must be lower than that of the wings' [7]. For our tail design an NACA 0012 was decided for both horizontal and vertical tails, an AR of 3-5 for the horizontal tail and an AR of 1-2 for the vertical tail was decided upon.

Control Surfaces

- Ailerons
- Elevators
- Rudders

The propulsion system plays an important role in an aircraft as it affects its thrust, efficiency, weight, and flight performance. Its main purpose is to provide enough thrust to overcome drag and meet the aircraft's flight requirements. The main choices in propulsion design for this project were the number of motors, number of blades, motor placement, and motor type

For our project we are constrained to choose between 2 or 4 motors. We ultimately decided upon 2 motors for our design due to the complexity that would come from choosing 4 motors. One of the reasons we chose 2 motors, besides the fact that most current general aircrafts only use 2 motors, is the power consumption that 4 motors would cause. Although we are free to

use any motor we want we are constrained by our battery, having 4 motors would lead to more draw from our battery and may cause issues during the flight demonstration of the competition. Furthermore there is an added complexity of having 4 motors; more motors would lead to more weight in our aircraft as well as a more complex wiring and electrical system with extra speed controllers.

For the number of blades per propeller we can choose between 2 or more blades. The advantages of having 2 blade setup is that they offer good thrust and energy consumption, while more blades lead to higher drag and require more torque, which is an issue as we are limited by our power supply. There are however advantages that would come from having more blades, such as high thrust at low speed, stability, control, and load-bearing capabilities [8]. We decided to go with a 3 blade propeller setup as it would provide some of the advantages that come from having a large number of propellers while not sacrificing the benefits that come with having 2 blades.

After selecting the number of motors and propeller blades, the next step is to determine the motors' positions on the wings. Motor placement can significantly affect airflow, lift distribution, and drag due to interactions between the propeller slipstream and the wing. The main configurations are tractor, where the propellers are positioned in front of the wing, and pusher, where they are positioned behind it. Propeller rotation direction can also influence aerodynamic performance. It was seen that a wingtip-mounted tractor configuration with inboard-up propeller rotation provided the highest aerodynamic efficiency by weakening the wingtip vortex and reducing induced drag. However, this configuration also increased the wing-root bending moment, potentially requiring a stronger and heavier wing. In comparison, a conventional tractor configuration with inboard-up rotation provided a good balance between aerodynamic efficiency and lower structural loads [12]. Based on these findings, we selected a conventional tractor configuration with inboard-up propeller rotation to maintain good aerodynamic performance while limiting additional structural loads.

The two main landing gear configurations considered were tail-gear and tricycle landing gear. Tail-gear configurations are simpler and can be lighter, but they provide less directional stability during ground operations and can be more difficult to control during takeoff and landing. In comparison, tricycle landing gear provides better ground stability and a more level fuselage attitude, making cargo loading easier. However, it requires an additional nose gear, which adds weight and mechanical complexity. We selected the tricycle landing gear configuration because its improved stability and ease of control during taxiing, takeoff, and landing make it better suited for our aircraft. Its level ground attitude also makes loading and transporting the required payload easier.

Power distribution in this aircraft consists of two individual circuits. The propulsion system and the control surfaces system. The propulsion system is restricted to a LiPo 14.8 V - 2200 mAh battery which drives the motor driven propellers using electronic speed controllers and an arming plug. The control surfaces system is restricted to a LiPo 1000 mAh battery that powers the receiver and the control surfaces.

The primary objective of maximizing payload capacity off a 100 foot runway dictates a high torque motor configuration paired with a low pitch - high static thrust propeller. This high load propulsion requirement directly governs the discharge current (C rating), thermal limits, and total capacity demands placed on the primary battery system.

Motor choices are narrowed down to three sizes: 2836, 3542, 4250. Where the first two numbers are the overall outer diameter of the motor rotor in millimeters and the second two numbers are the overall length of the motor case in millimeters. Main considerations for motor sizes are mass and internal heat loss. Smaller batteries are lighter but under higher power loads the internal heat dissipation will cause the motor to overheat and fail. Larger batteries are better at heat dissipation but their mass calls for higher power which our 2200 mAh will struggle to provide.

Furthermore motor selection is contingent on its kV [RPM/Volt] value, or speed constant. This is a good indicator of speed to torque ratios for each motor. The higher the kV value the higher the top rpm of the motor, but the lower the torque, and vice versa for lower kV values.

Additionally, the power distribution is driven by propeller blade choice. Where our propellers will be at a maximum of 12”, although propeller pitch is an important factor. Where a lower pitch acts as a lower gear, it produces higher torque at lower speeds. Where each option’s value is a representation of its total forward movement in inches in theoretically perfect conditions.

These three factors help shape the C value for the 2200 mAh battery, or 2.2 Ah battery. Our plane only needs to operate in the order of minutes as opposed to hours. The C rating is a maximum power delivery constant that helps identify the maximum amps the battery can output at a given time. Higher C ratings have higher mass, but higher C ratings allow more amps to leave the battery, providing more power to the propellers. Although excessive C ratings give diminishing returns, more weight added to the system calls for more power, which the battery cannot provide for as long due to its limited capacity.

Battery and electronics are still to be determined in the control surface circuit.

Avionics

Battery

Other stuff

Chapter VI: Description of Project Design

Fuselage Design:

The plane's fuselage must house all necessary electronics and additional payload, provide a structural base for the wings, tail, and landing gear, and streamline aerodynamics. Our philosophy is an inside-out design, meaning the payload and inner components drive the design for the fuselage.

In general, the fuselage will have a rectangular box shape to house the electronics and payload. Storage for electronics are put in the front, with the payload behind. If the payload were to be put too far in the front of the plane, adding any bottles to the cargo bay would significantly change the location of the Center of Gravity and have drastic effects on the plane's stability. In front of everything, the nose sits with a length of 15 inches for aerodynamic streamlining and has a faceted wedge shape that is flat at the front. It is important to note that sharp vertices will be sanded down to triangular filets to avoid the significant drag caused by square corners [9]. Access to the cargo bay will be through a ramp door from behind. The electronics will be accessible via removable panels. The tail arm of the aircraft will consist of an aluminum rod with a starting length of 60 inches, however this length is subject to change. This turns out to be a 115 inch fuselage, keeping within the 120 inch limit.

For the cargo bay, the payload will account for up to 4 bottles, all laid out flat to make each bottle easy to secure. This cargo bay has a planned size of 10 inches wide and 26 inches long, with a height of at least 5 inches to account for our payload layout. Also, placing the cargo bay at the back means it can be accessed through a rear-facing drop-down door. This door was selected due to its ease of manufacture and securing, and its limited interaction with other parts. The door would lower with a piano hinge to guide its movement and a hook-and-eye latch at the top to secure the door when closed. Within the cargo bay, bottles will be held down with Velcro straps that attach to hooks on the floor of the payload area.

To gain access to the payload, a ramp door behind the cargo bay will open downward. A major design consideration for this door was the upsweep angle. This must be small enough for the door to not interfere with any aerodynamics for the aircraft itself. If the angle were to be too large, then the sudden decrease in cross sectional area would result in a low pressure area that would pitch the nose up and induce parasitic drag [2]. Testing reports that an ideal upsweep angle is between 10°-14° [10]. However the drag produced is often overestimated due to wing downwash–especially from high-mounted wings–and constrained by ground clearance on takeoff [9]. Therefore, we decided on a door that is 15 inches long, creating an angle of 18.43°.

Fineness ratio (F) was an important design consideration for this fuselage design and is defined as the ratio comparing the overall length (L) of the fuselage relative to its equivalent diameter (D). This relationship can be described by the following equation:

$$F = \frac{L}{D}$$

Ideally, this ratio falls in between 6 - 8 to minimize the parasitic drag induced by the flow separation of a blunt body and parasitic skin friction [9]. The overall length for this fuselage (not

including the 60 inch tail arm) is $L = 56$ inches, and the equivalent diameter is $D = 7.979$ inches. This results in a fineness ratio of $F = 7.019$.

As with the majority of the aircraft, the skeleton will be assembled from a reinforced balsa wood with a mylar film wrapped around the frame to create a skin. There will be a floor for the electronics and cargo bay sections that is also constructed from balsa wood so that components and payload can be supported. The walls and ceiling will be in the shape of a Warren truss such that the triangular framework can disperse axial load throughout the airframe, as common in aircraft. Another benefit for this selection is its geometric repeatability for manufacturing ease and material efficiency [11].

Wing Design and Tail Design:

Using the ratios above we are able to create a calculator to help us select our AR, Taper Ratio, and maximize wing area while maintaining a legal, manufacturable aircraft.

Taper Ratio	AR	Wingspan (ft)	Wingspan (in)	Mean Chord (in)	Tip Chord	Root Chord	Wing Area (in^2)
0.45	8.0	8	96	12.000	7.448	16.552	1152.000
0.50	8.0	8	96	12.000	8.000	16.000	1152.000
0.55	8.0	8	96	12.000	8.516	15.484	1152.000
0.60	8.0	8	96	12.000	9.000	15.000	1152.000
0.65	8.0	8	96	12.000	9.455	14.545	1152.000
0.70	8.0	8	96	12.000	9.882	14.118	1152.000
0.75	8.0	8	96	12.000	10.286	13.714	1152.000
0.80	8.0	8	96	12.000	10.667	13.333	1152.000
0.45	7.0	8	96	13.714	8.512	18.916	1316.571
0.50	7.0	8	96	13.714	9.143	18.286	1316.571
0.55	7.0	8	96	13.714	9.733	17.696	1316.571
0.45	6.0	8	96	16.000	9.931	22.069	1536.000
0.50	6.0	8	96	16.000	10.667	21.333	1536.000
0.55	6.0	8	96	16.000	11.355	20.645	1536.000
0.60	6.5	8	96	14.769	11.077	18.462	1417.846

Wing dim
Wing AR
Wing airfoil
Wing config
Motor placement on wing
Tail dim
Tail config
Tail AR
Tail airfoil

Control Surfaces:

Aileron
Elevator
Rudder

Propulsion:

Motor choice

Landing Gear Design:

Design

Controls

Power Distribution:

A propeller pitch of 4.5 is chosen for now as it is one of the lowest pitch values available for a 12” diameter propeller.

A 3542 motor is chosen as a compromise across weight, heat dissipation, and amp draw.

A 800 kV variation is the lowest velocity constant motor available allowing the static thrust to be high enough to bring the plane in the air under limited runway space while still drawing an appropriate amount of power that allows the plane to make its flight and landing.

A 35 C battery is chosen as a compromise between weight and system safety margin for output current.

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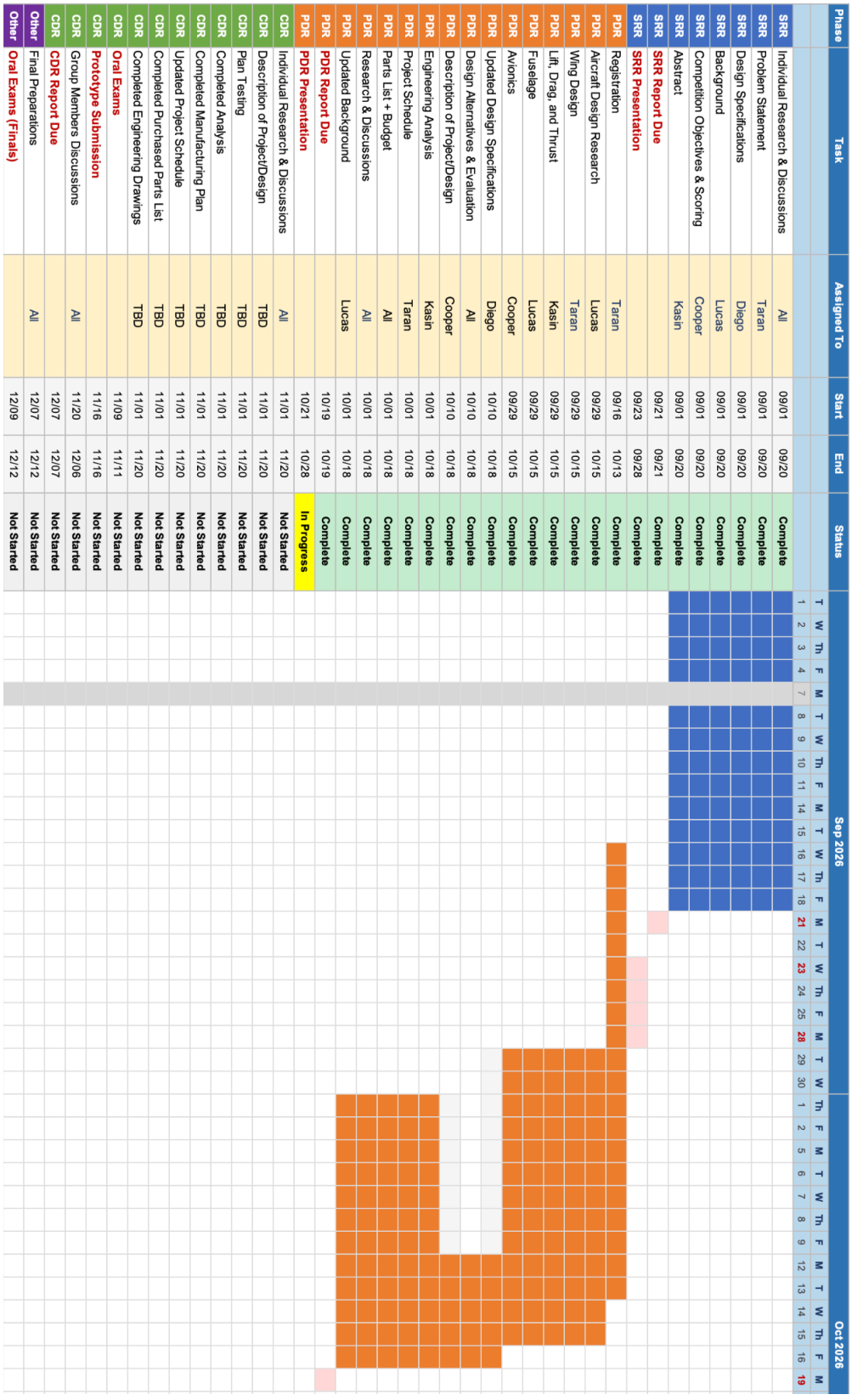
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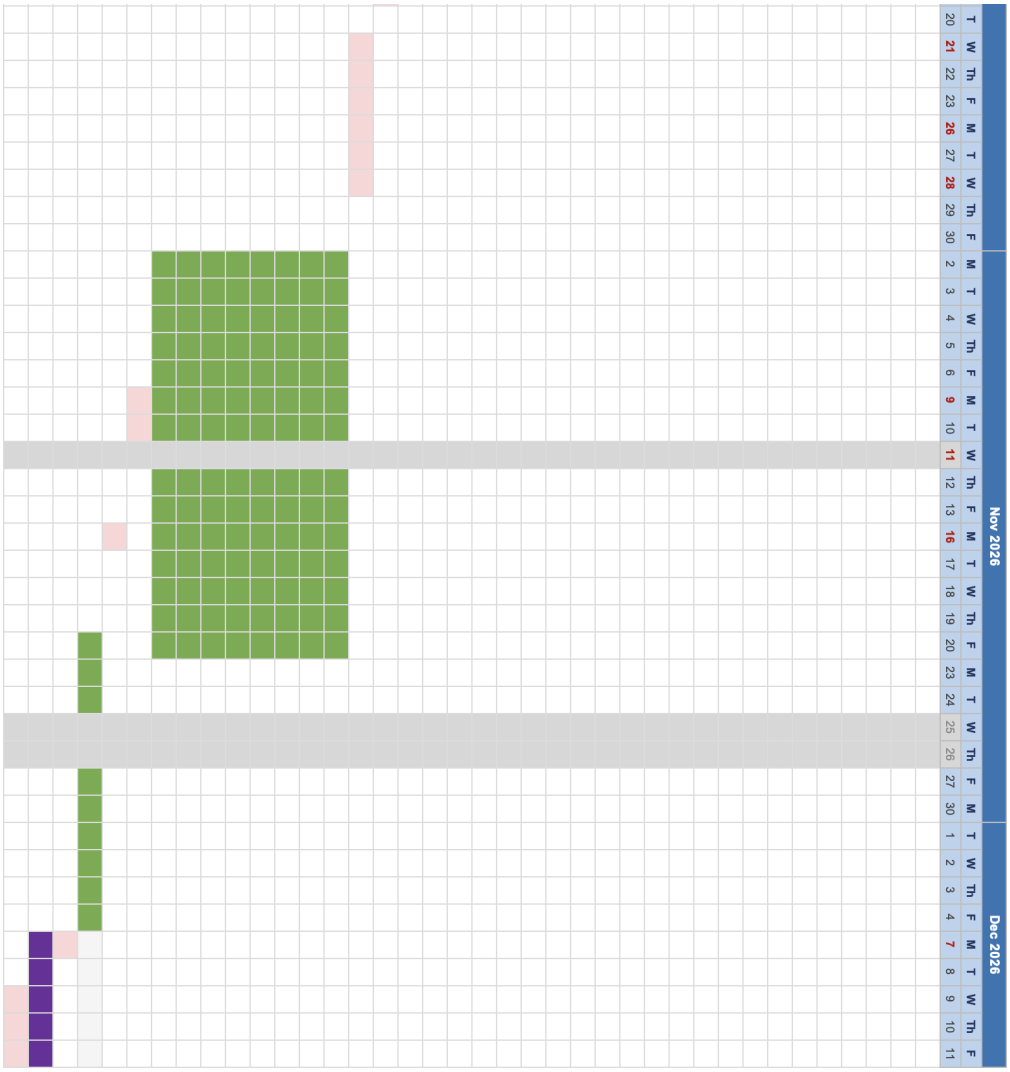
Appendix

A. Analysis

C. Project Schedule



Phase	Task	Assigned To	Start	End	Status
SRR	Individual Research & Discussions	All	09/01	09/20	Complete
SRR	Problem Statement	Taran	09/01	09/20	Complete
SRR	Design Specifications	Diego	09/01	09/20	Complete
SRR	Background	Lucas	09/01	09/20	Complete
SRR	Competition Objectives & Scoring	Cooper	09/01	09/20	Complete
SRR	Abstract	Kasin	09/01	09/20	Complete
SRR	**SRR Report Due**		09/21	09/21	Complete
SRR	**SRR Presentation**		09/23	09/28	Complete
PDR	Registration	Taran	09/16	10/13	Complete
PDR	Aircraft Design Research	Lucas	09/29	10/15	Complete
PDR	Wing Design	Taran	09/29	10/15	Complete
PDR	Lift, Drag, and Thrust	Kasin	09/29	10/15	Complete
PDR	Fuselage	Lucas	09/29	10/15	Complete
PDR	Avionics	Cooper	09/29	10/15	Complete
PDR	Updated Design Specifications	Diego	10/10	10/18	Complete
PDR	Design Alternatives & Evaluation	All	10/10	10/18	Complete
PDR	Description of Project/Design	Cooper	10/10	10/18	Complete
PDR	Engineering Analysis	Kasin	10/01	10/18	Complete
PDR	Project Schedule	Taran	10/01	10/18	Complete
PDR	Parts List + Budget	All	10/01	10/18	Complete
PDR	Research & Discussions	All	10/01	10/18	Complete
PDR	Updated Background	Lucas	10/01	10/18	Complete
PDR	**PDR Report Due**		10/19	10/19	Complete
PDR	**PDR Presentation**		10/21	10/28	In Progress
CDR	Individual Research & Discussions	All	11/01	11/20	Not Started
CDR	Description of Project/Design	TBD	11/01	11/20	Not Started
CDR	Plan Testing	TBD	11/01	11/20	Not Started
CDR	Completed Analysis	TBD	11/01	11/20	Not Started
CDR	Completed Manufacturing Plan	TBD	11/01	11/20	Not Started
CDR	Updated Project Schedule	TBD	11/01	11/20	Not Started
CDR	Completed Purchased Parts List	TBD	11/01	11/20	Not Started
CDR	Completed Engineering Drawings	TBD	11/01	11/20	Not Started
CDR	**Oral Exams**		11/09	11/11	Not Started
CDR	**Prototype Submission**		11/16	11/16	Not Started
CDR	Group Members Discussions	All	11/20	12/06	Not Started
CDR	**CDR Report Due**		12/07	12/07	Not Started
Other	Final Preparations	All	12/07	12/12	Not Started
Other	**Oral Exams (Finals)**		12/09	12/12	Not Started



D. Purchased Parts List

F. Project Budget